# Ling 285
# Human Language and Technology
# Lecture 4

# Dynamical Systems 1

Khalil Iskarous

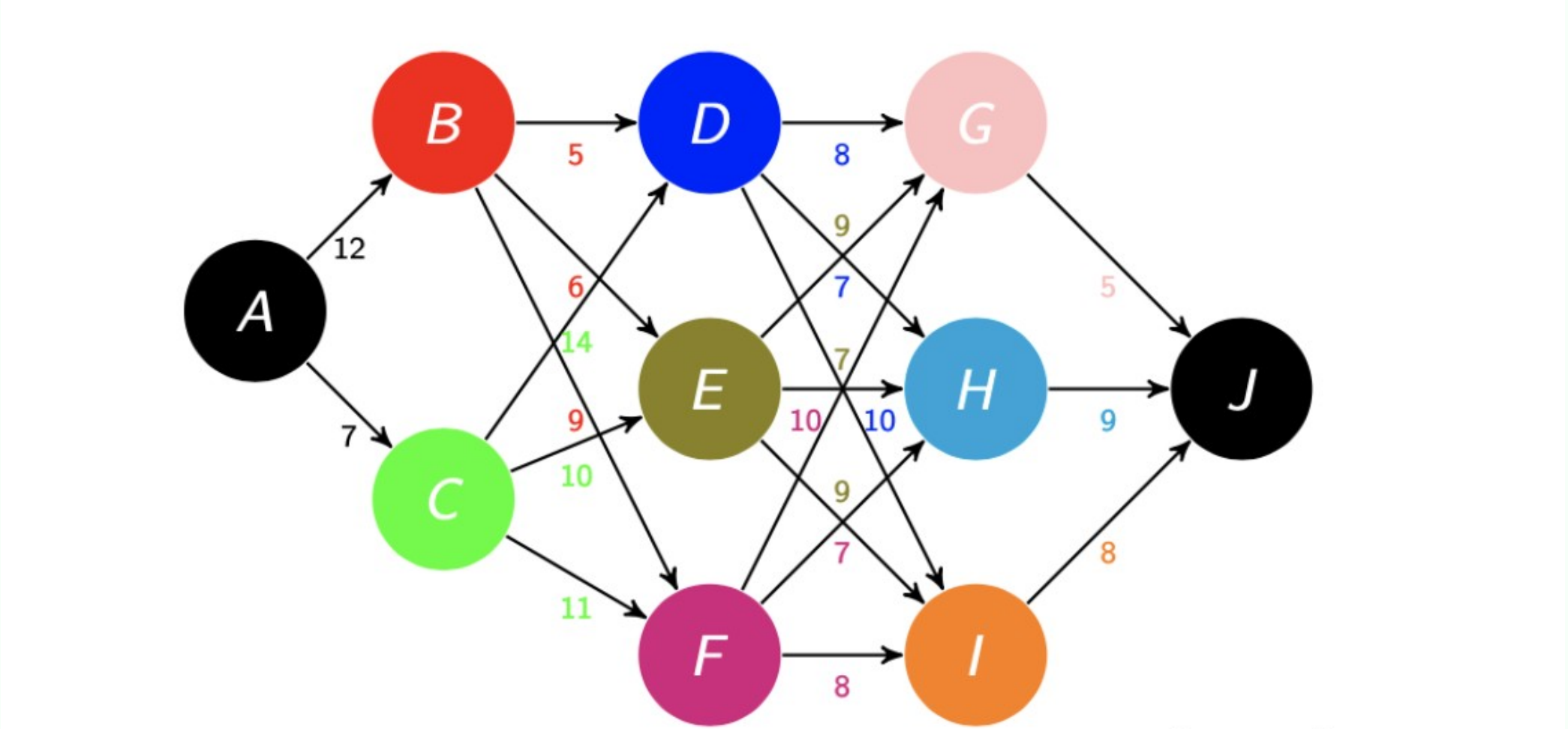
Department of Linguistics

University of Southern California

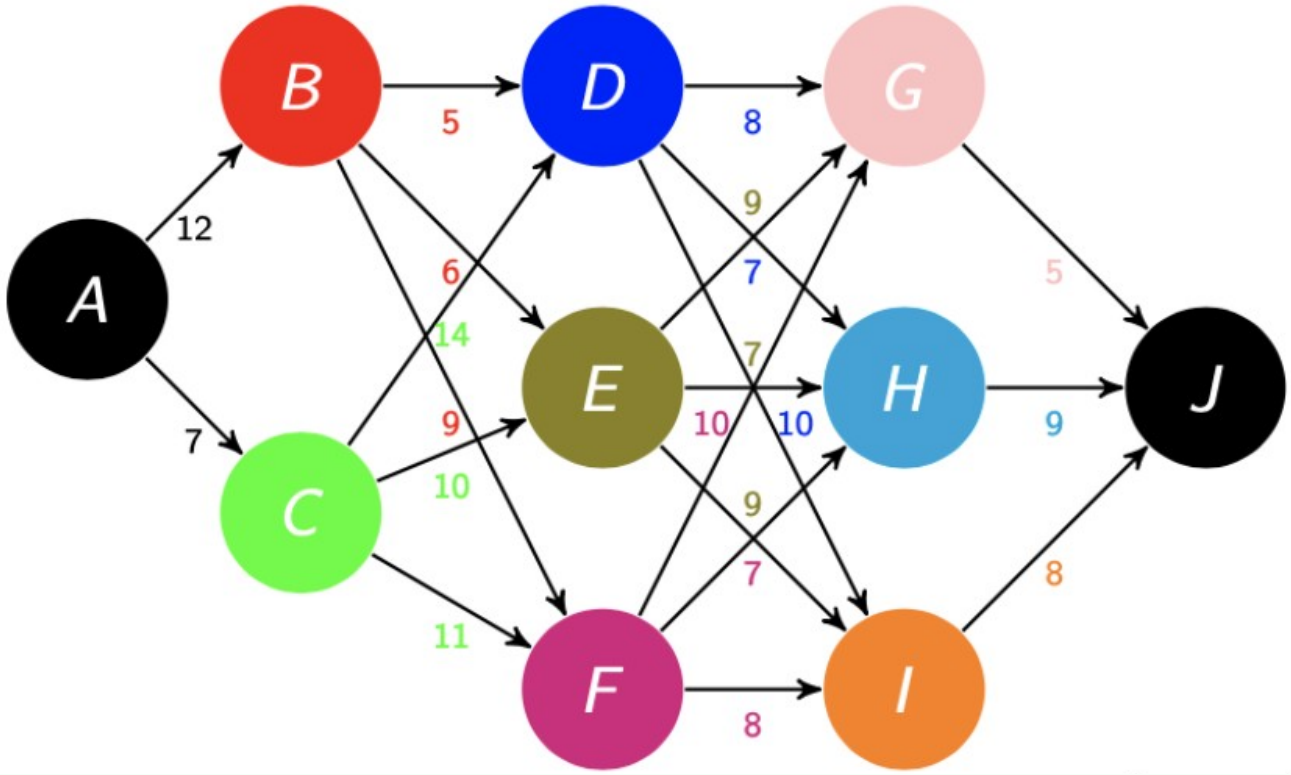
# Today

- Dynamic Programming
- Pop Quiz
- Dynamical Systems Galore

# Shortest Path Problem

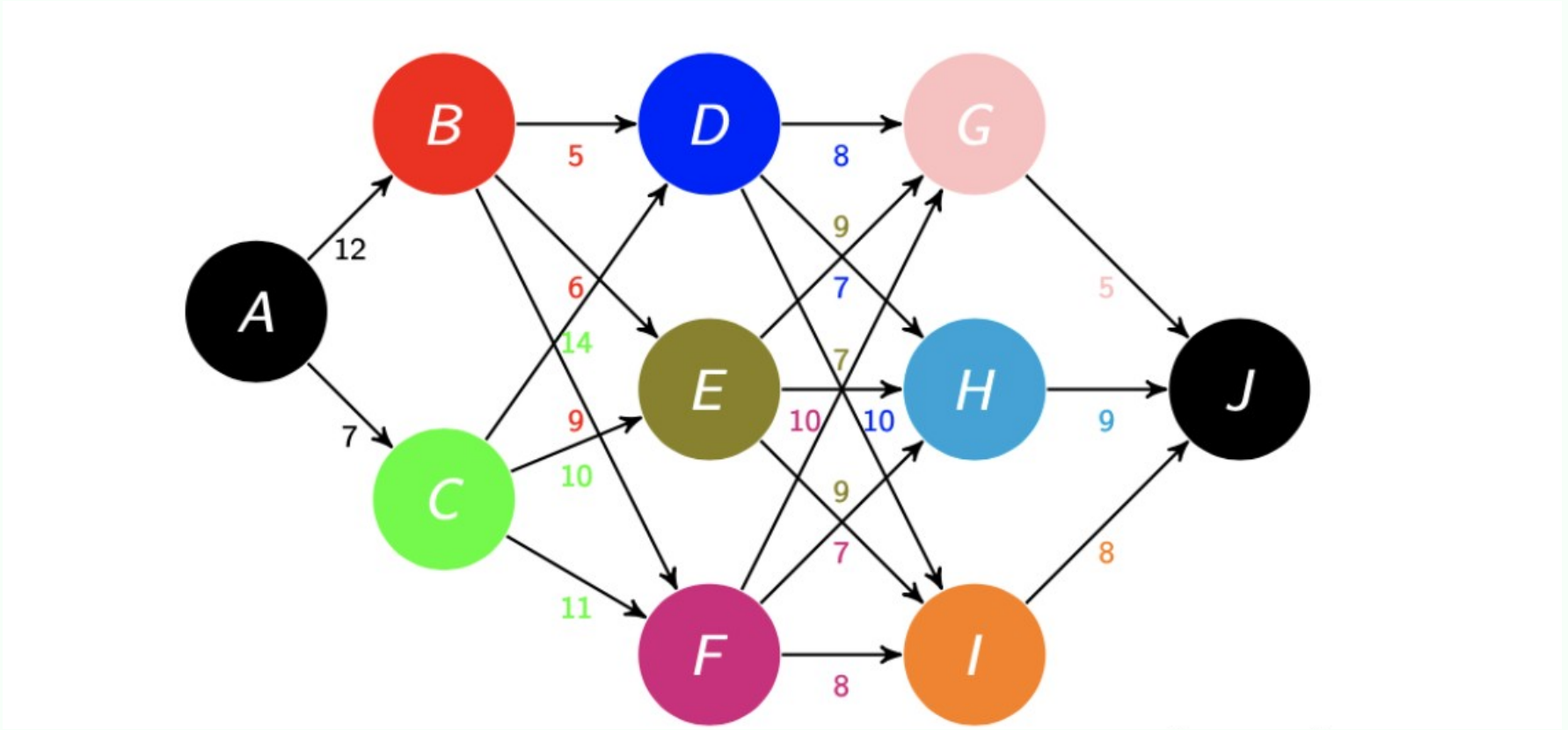


# Greedy Algorithm



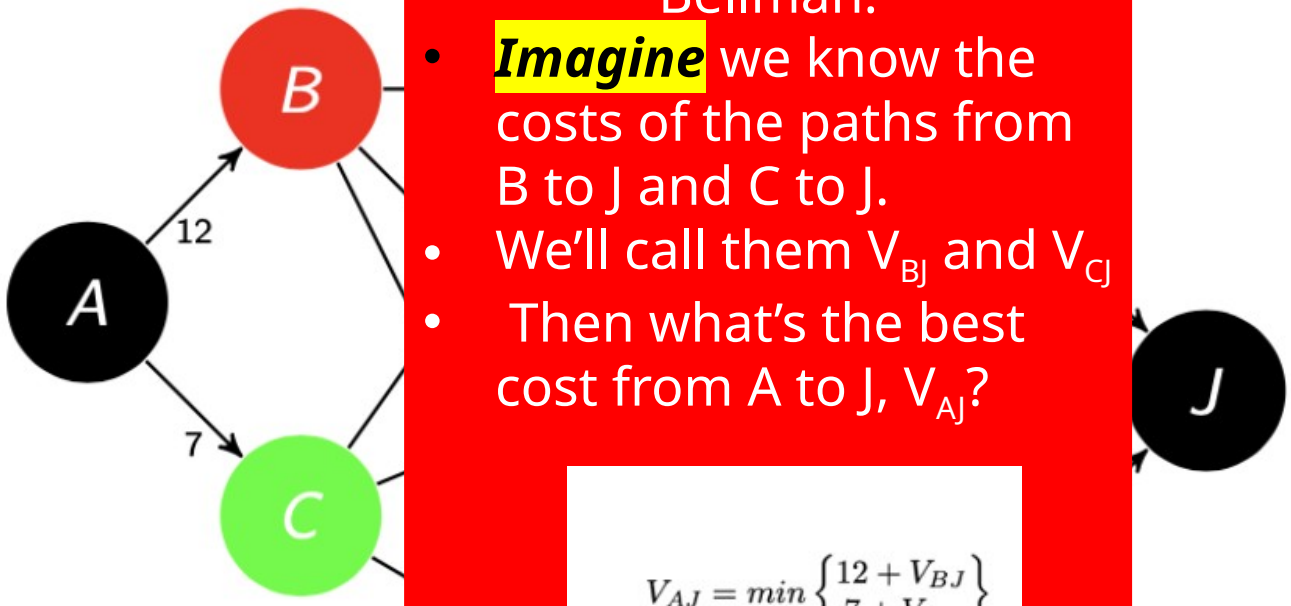

News Flash: Greed is not good!!

# Exhaustive Algorithm




Very Accurate, but if there are many *stages* and *states, it will take a very very long*

# Dynamic Programming

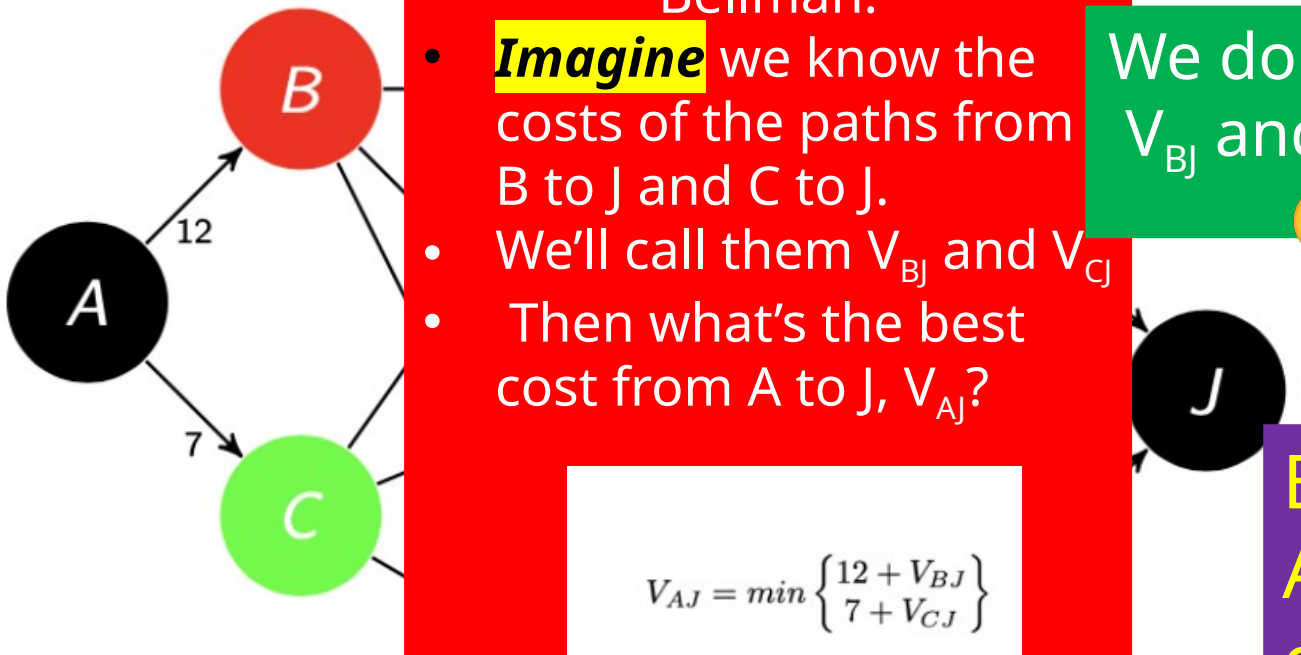



Bellman:

- ***Imagine*** we know the costs of the paths from B to J and C to J.
- We'll call them $V_{BJ}$ and $V_{CJ}$
- Then what's the best cost from A to J, $V_{AJ}$?

$$V_{AJ} = min \begin{Bmatrix} 12 + V_{BJ} \\ 7 + V_{CJ} \end{Bmatrix}$$

# Dynamic Programming




Bellman:

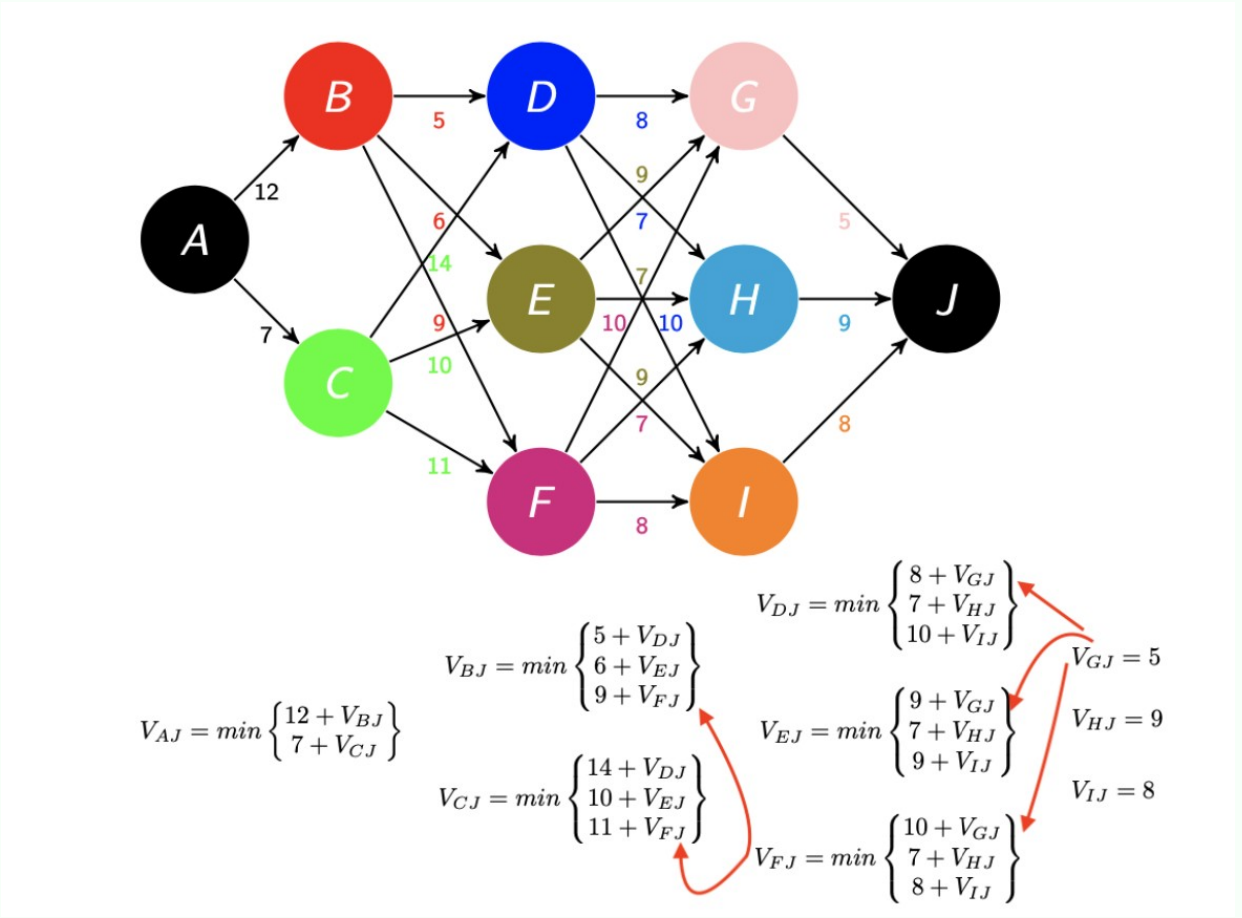
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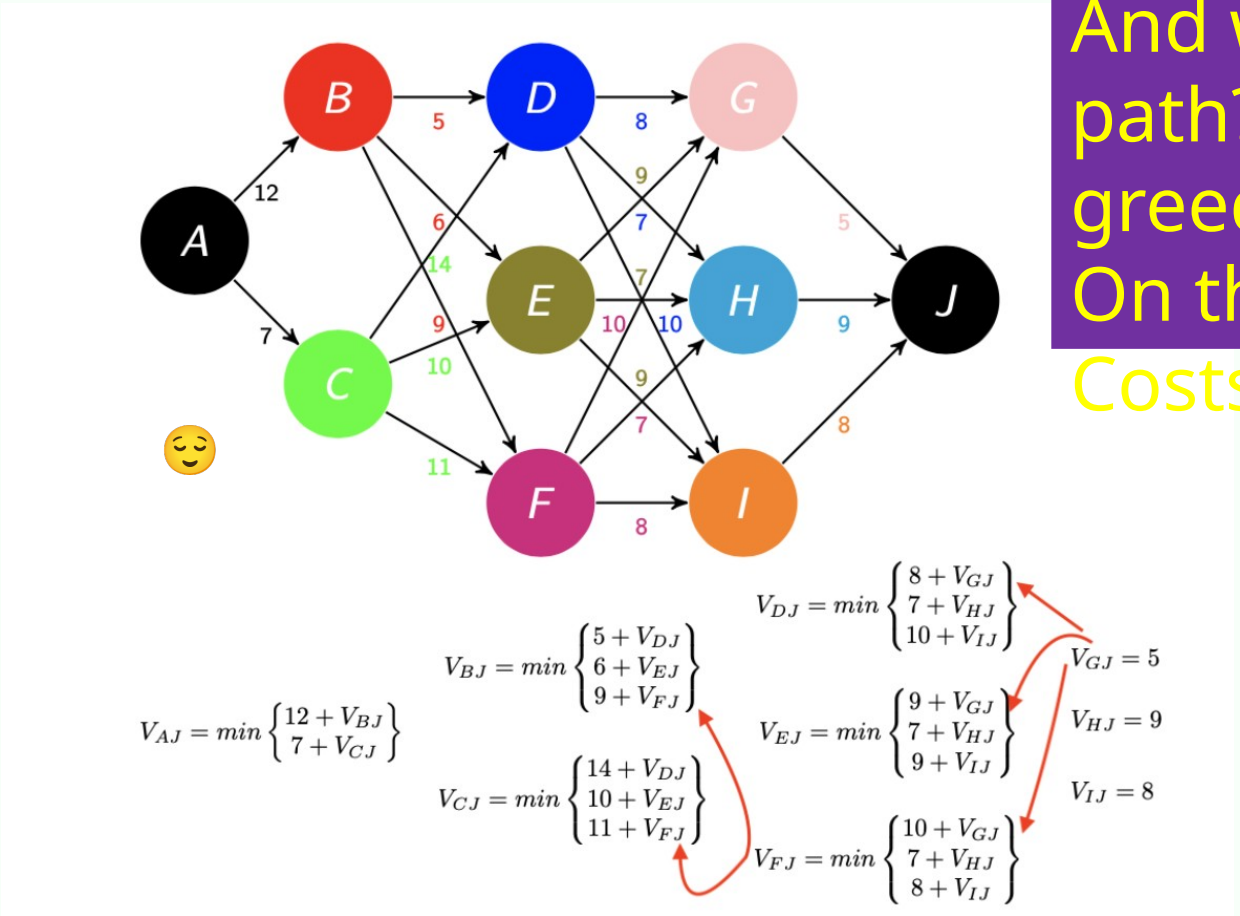
We don't know $V_{BJ}$ and $V_{CJ}$ ☹ ☹

Bellman: To solve A problem, decompose it into subproblems You can solve, *if you can*!

Bellman: We don't know the best cost from B to J, but if we knew the best costs from D, E, F, to J, then we can find the best cost from B to J



# Dynamic Programming: Calculate Values, then you can be greedy!

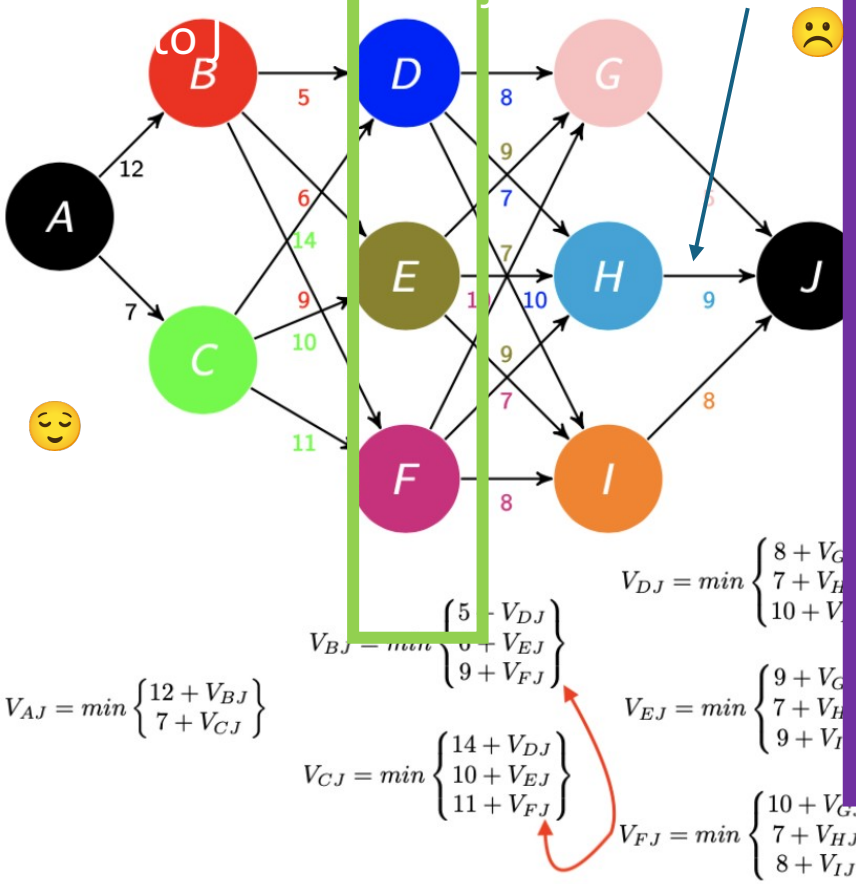



And what's the best path? Now go greedy
On the Values! Not Costs!!!!

# Dynamic Programming is forsight!

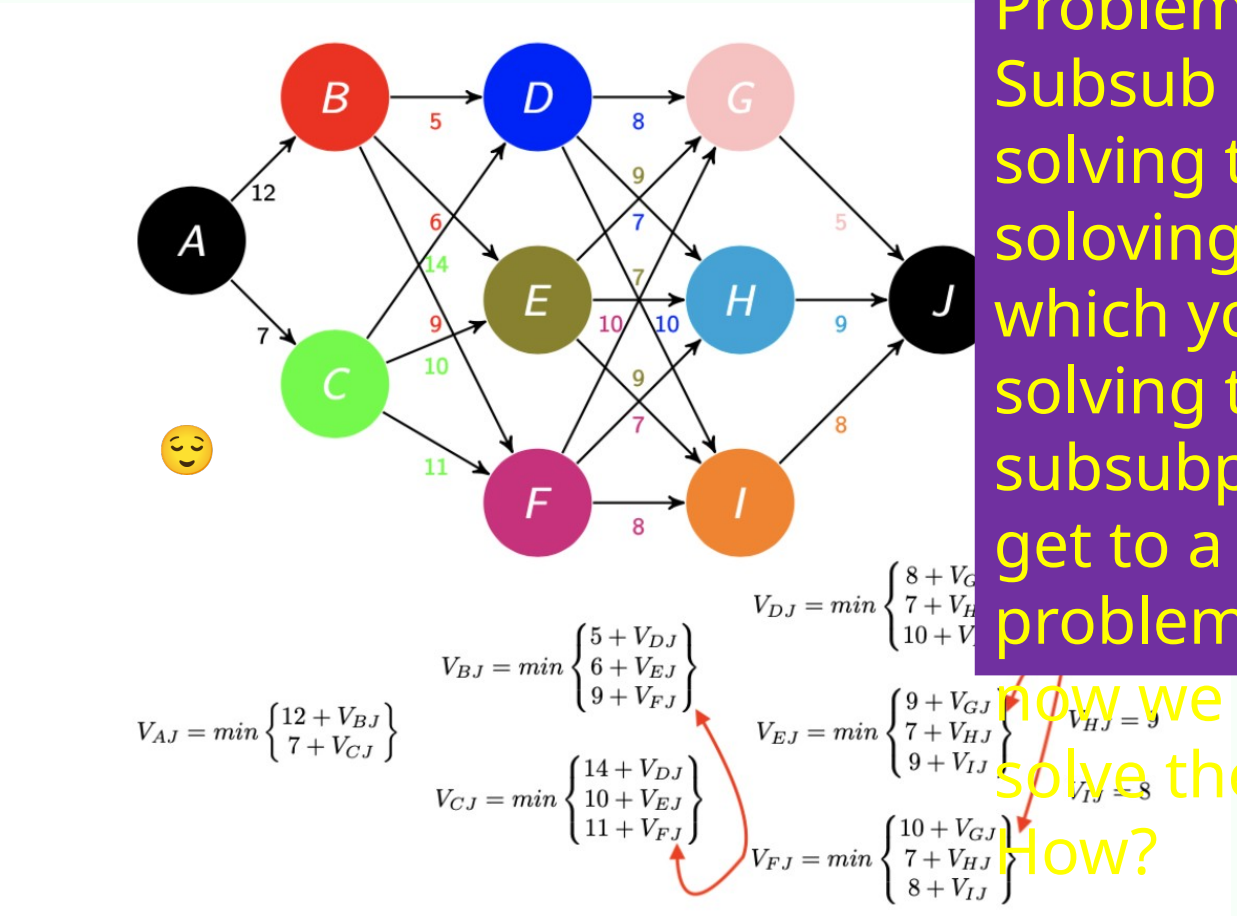
**Søren Kierkegaard: Life can only be understood backwards; but it must be lived forwards**

🤔If I'm at this stage, DP gives me forsight that State H leads to something terrible 🤨 This is still the case if




In the initial problem, a late obstacle (the 9!) can lead an early greedy decision based on the cost to lead to a bad path. The Values contain the FUTURE!!!! So we can go greedy on them, since they already contain info on all the future obstacles!!! So they make present decisions optimal, because the whole future from each state is encapsulated in its Value!!!

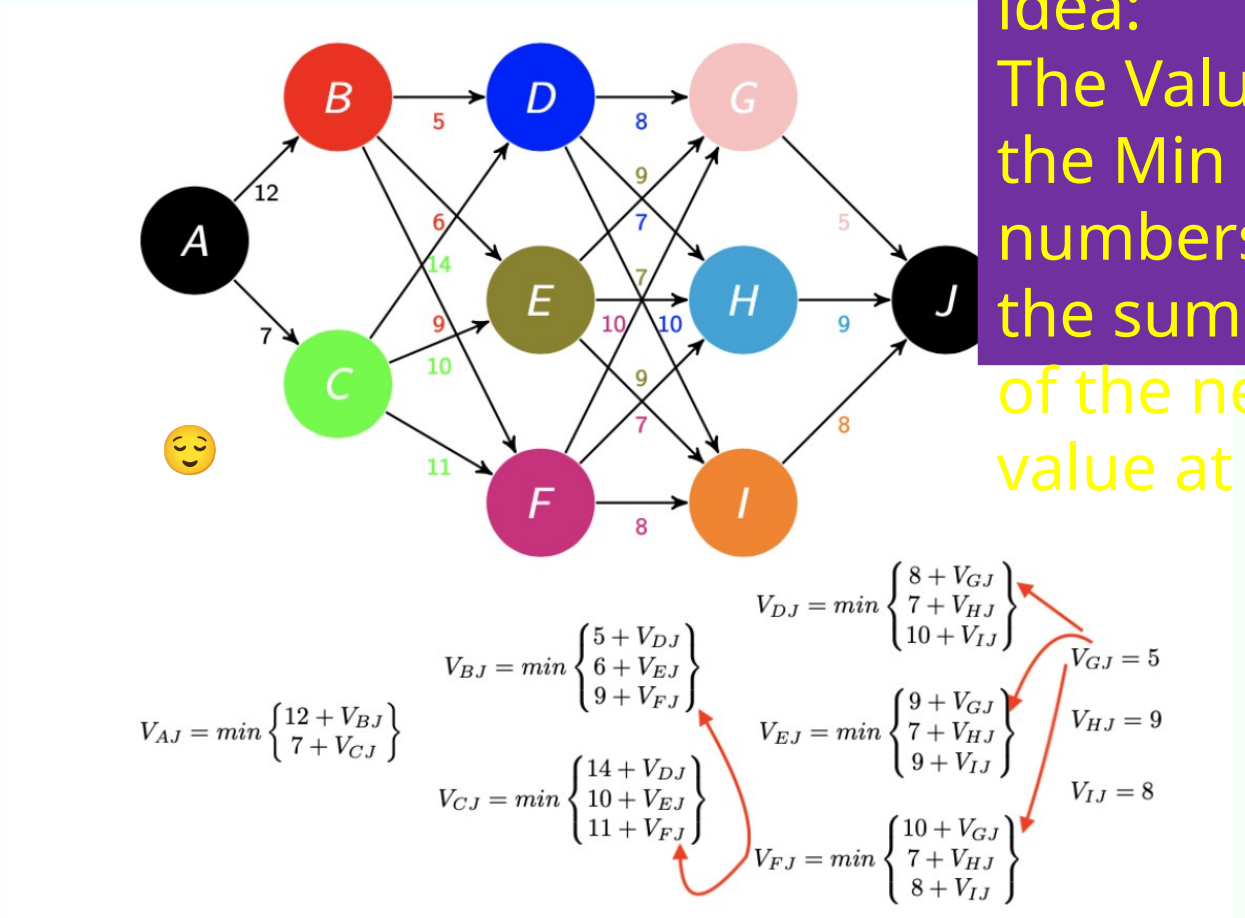
# Dynamic Programming:



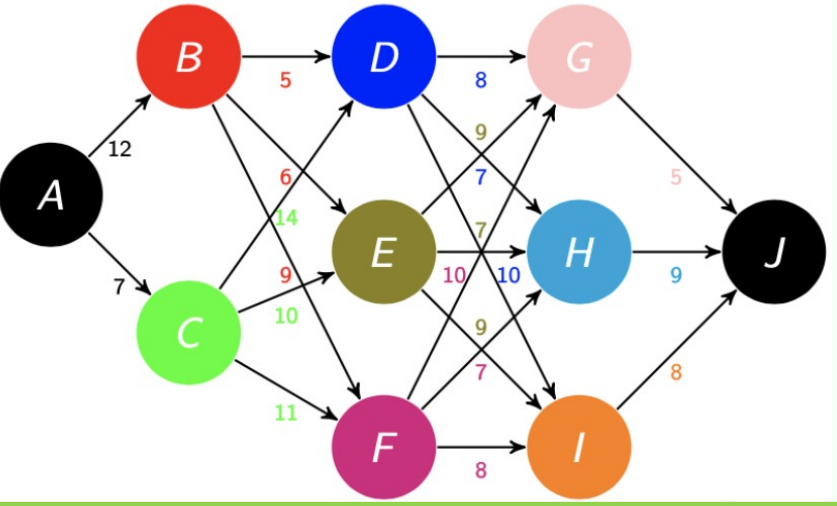

Dynamic Programming can be seen to start with posing a Problem, its subproblems, Subsub Problems…. Then solving the big problem by soloving the subproblems, which you can do by solving the subsubproblems…. Till you get to a point, where the problems are *trivial*, and now we can go back and solve the bigger problems. How?

# Dynamic Programming:

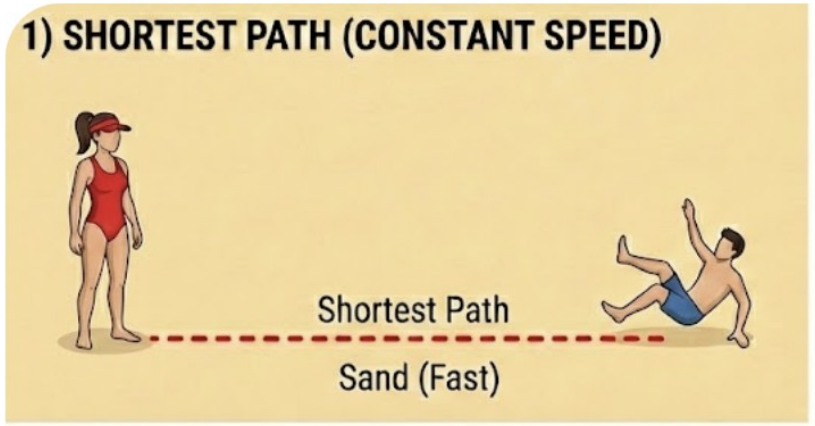
By *propagating* Values from each stage to the earlier stage by using Bellman's idea:
The Value at one state is the Min of a set of numbers, each of which is the sum of the cost to one of the next states and the value at that state!

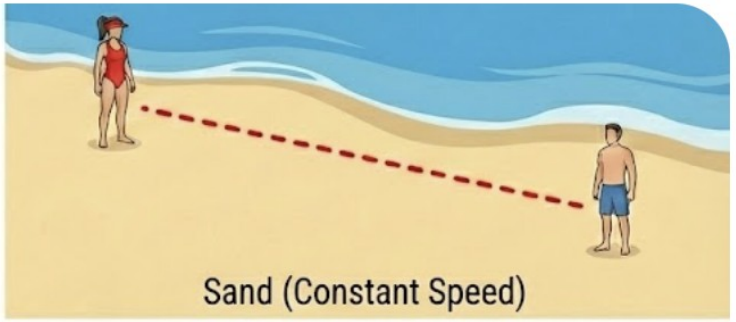


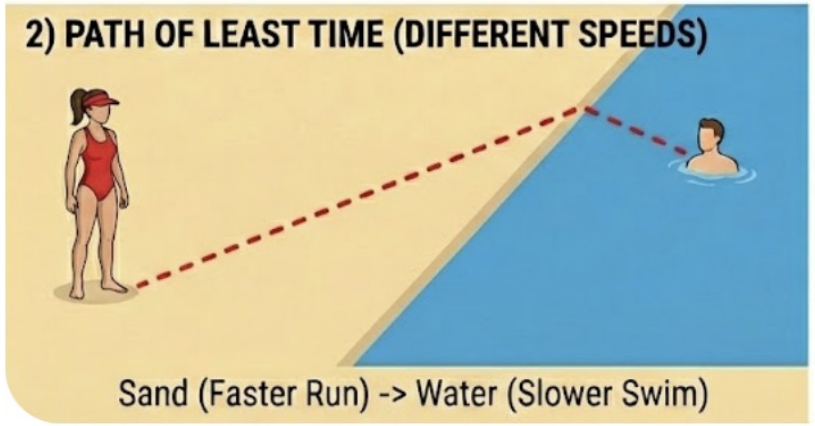
# Mind-Blowing Stuff




**DP: Go through the bad 12 earlier**
**to get onto a route that does**
**get stuck later at the 9! It ma**
seem
that we need a big brain to do
that!!!

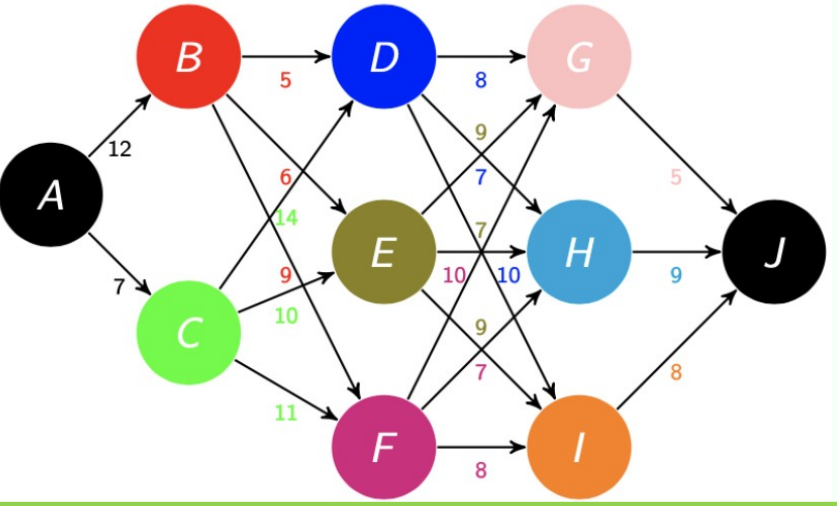


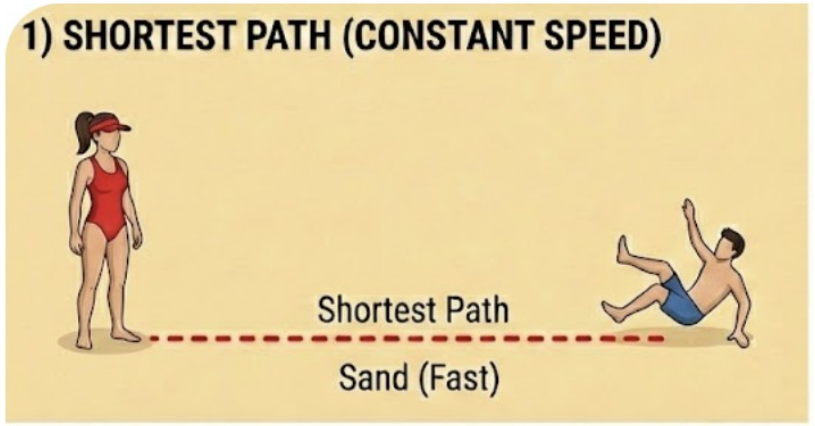



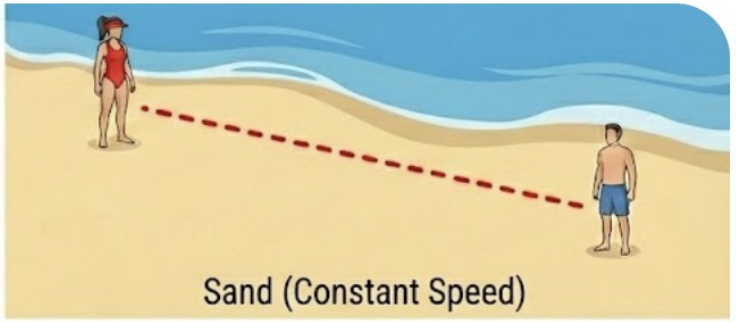

Nano Bannana Pro

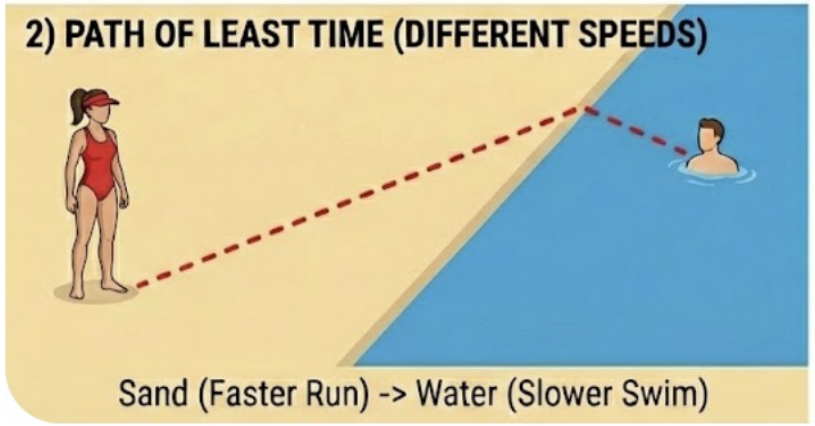
# Decision-Making is like Physics ☺




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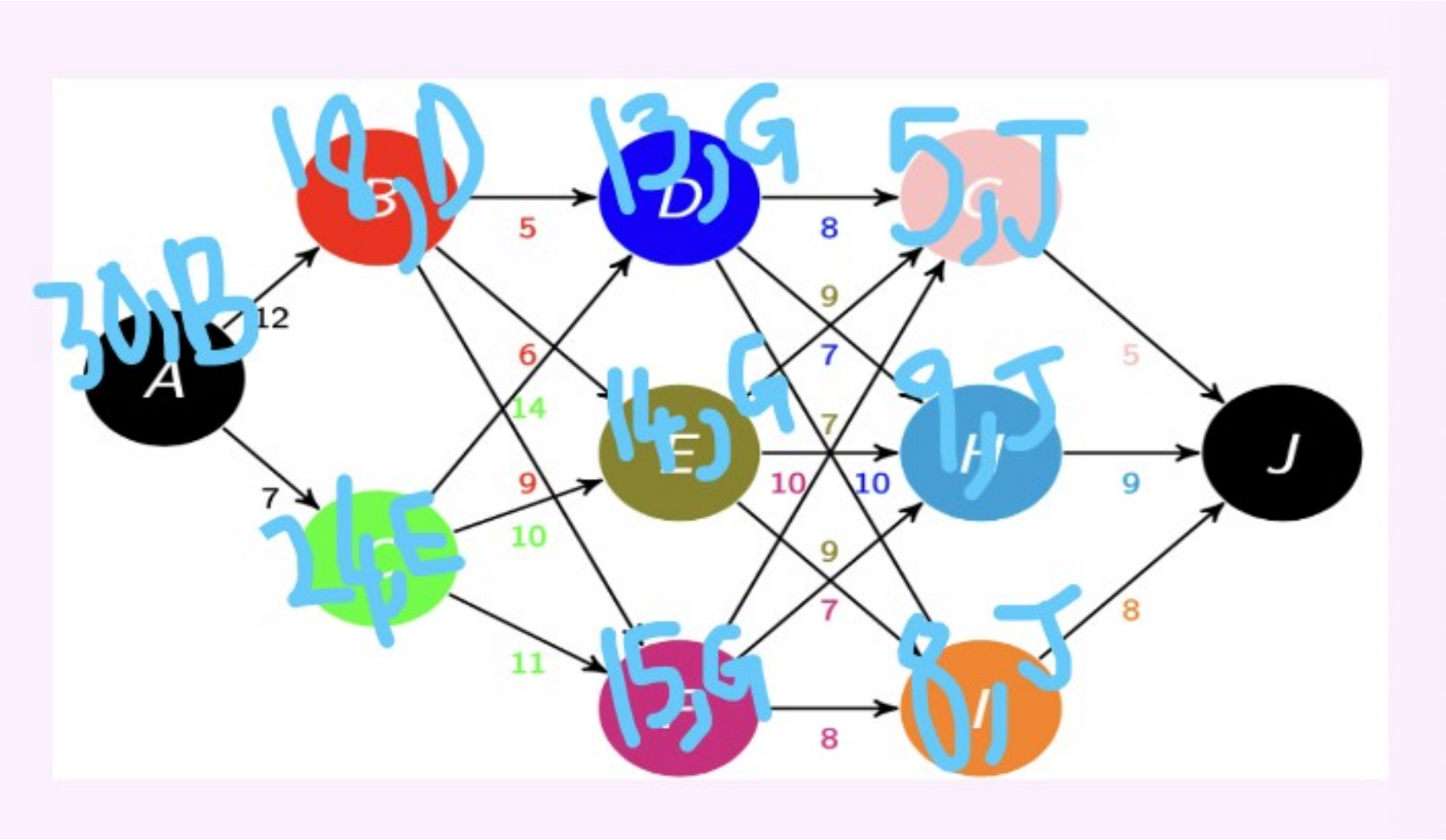


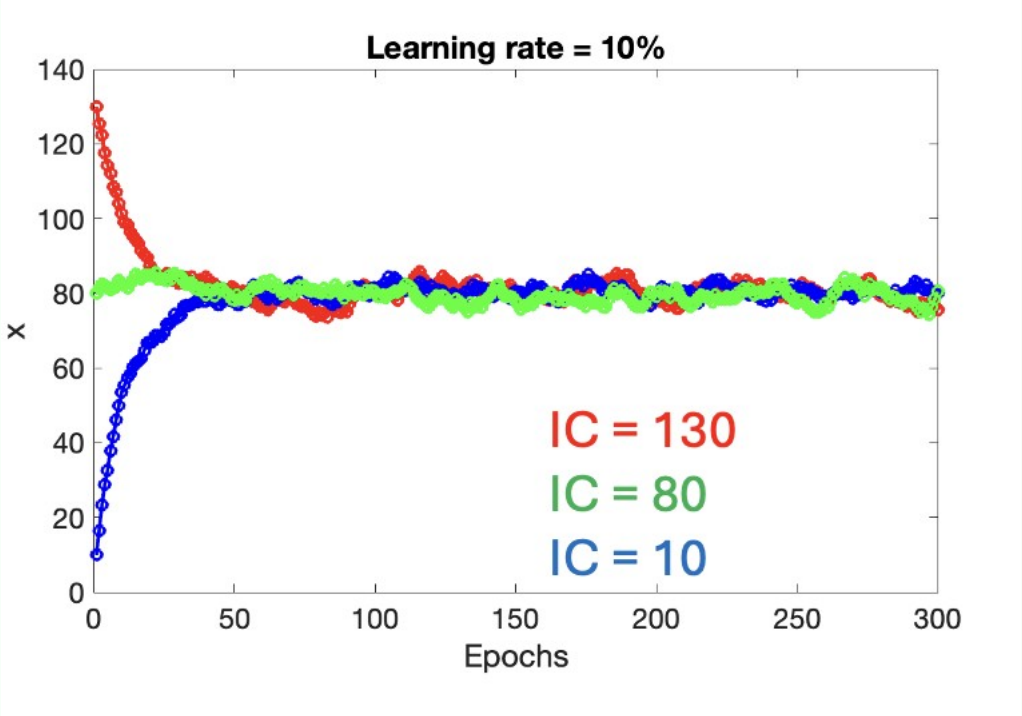


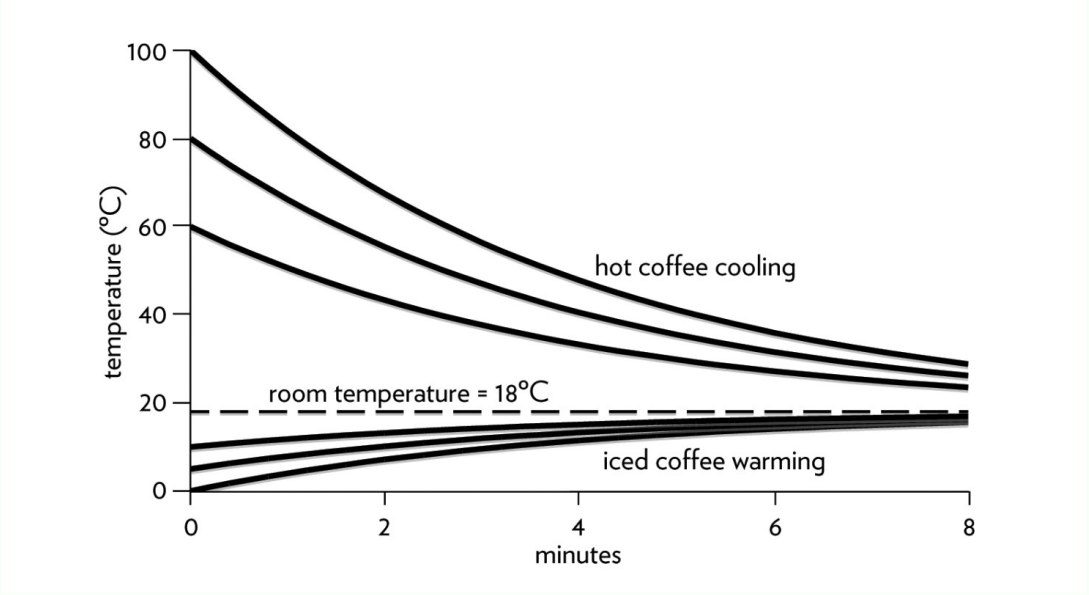

# Figuring out the best Route!



# Pop Quiz

# Pop QUiz


Learning rate = 10%
x
Epochs
IC = 130
IC = 80
IC = 10


temperature (°C)
minutes
hot coffee cooling
room temperature = 18°C
iced coffee warming